251201 Fourier Transform

Video:

- 2:30 (After Paint Colors) Pause and Motivate decomposing TD Signals into FD for TFA.

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ex./ $f(t) = \delta(t)$

$$\Rightarrow \tilde{F}(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$\delta(t)$ is non-zero only for $t=0$

$$\Rightarrow \tilde{F}(\omega) = e^{-j\omega(0)} \quad \boxed{\therefore \tilde{F}(\omega) = 1}$$

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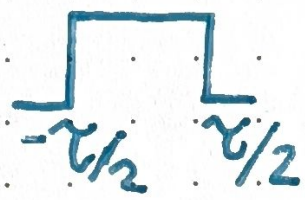
ex./ $f(t) = \text{rect}(t/\tau)$



$$\tilde{F}(\omega) = \int_{-\infty}^{\infty} \text{rect}(t/\tau) e^{-j\omega t} dt = \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt$$

$$\Rightarrow \tilde{F}(\omega) = -\frac{e^{-j\omega t}}{j\omega}\Bigg|_{-\tau/2}^{\tau/2} = j\frac{e^{-j\omega\tau/2}}{\omega} - j\frac{e^{j\omega\tau/2}}{\omega}$$

$$= \frac{j}{\omega}\left(e^{-j\omega\tau/2} - e^{j\omega\tau/2}\right)$$

Use: $\sinh(x) = (e^{x} - e^{-x})/2$

$$\Rightarrow \tilde{F}(\omega) = \frac{2j}{\omega} \sinh\left(\frac{-j\omega\tau}{2}\right)$$

Now Use: $\sinh(x) = -j\sin(jx)$

$$\Rightarrow \tilde{F}(\omega) = -\frac{2j}{\omega} j \sin\left(-j^2 \omega \tau/2\right) = \frac{2}{\omega}\sin\left(\frac{\omega\tau}{2}\right)$$

$$\Rightarrow \tilde{F}(\omega) = \tau \frac{\sin(\omega\tau/2)}{\omega\tau/2}$$

Now Use: $\mathrm{Sinc}(x) = \sin(x)/x$ where $\mathrm{Sinc}(0) = 1$

$$\boxed{\therefore \tilde{F}(\omega) = \tau\, \mathrm{Sinc}(\omega\tau/2)}$$

ex./ $f(t) = A\cos(\omega_0 t + \phi_0)$

$$\Rightarrow \tilde{F}(\omega) = \int_{-\infty}^{\infty} A\cos(\omega_0 t + \phi_0)\, e^{-j\omega t}\, dt$$

Use: $\cos(x) = (e^{jx} + e^{-jx})/2$

$$\Rightarrow \tilde{F}(\omega) = \frac{A}{2}\int_{-\infty}^{\infty} \left(e^{j(\omega_0 t + \phi_0)} + e^{-j(\omega_0 t + \phi_0)}\right) e^{-j\omega t}\, dt$$

$$= \frac{A}{2} e^{j\phi_0} \int_{-\infty}^{\infty} e^{jt(\omega_0 - \omega)} + e^{-jt(\omega_0 + \omega)}\, dt$$ Looks like 2 FTs

Use: $FT[f(t) = 1] \Rightarrow \tilde{F}(\omega) = 2\pi\delta(\omega)$

$$\Rightarrow \tilde{F}(\omega) = \frac{Ae^{j\phi_0}}{2}\left(2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0)\right)$$

$$\boxed{\therefore \tilde{F}(\omega) = \pi A e^{j\phi_0}\left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\right)}$$